Frederic Rosen, Algebra of Mohammed Ben Musa (1831)

Islamic Original

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

الحساب علم مبادي شرح

الظنون كشف
الابي ابن
والمقابلة الجبر في مسلم بيد صنفه كتاب اول هذا

فصحبه بغداد. الى الروم بلاد من راجعا موسى بن محمد عليه قدم ان الى بها فاقام كفرطابا وسكن حران من خرجقرة بن ثابت
الحجاب اهل جملة من وكان المتوكل الى وقربه داره فنزله بغداد الى فحمله وذكاءه فضله وعرف معرفته في واجتهد
المزيحفي

$1^{st} cx^2 + bx = a$
$2^d cx^2 + a = bx$
$3^d cx^2 = bx + a$
$1^{st} cx^2 + bx = a$
$x^2 + 10x = 39$
$x = \sqrt{25 + 39} - 5 = \sqrt{64} - 5 = 8 - 5 = 3$

$cx^2 + bx = ax^2 + \frac{b}{c}x = \frac{a}{c}$
$2x^2 + 10x = 48x^2 + 5x = 24$
$x = \sqrt{6\frac{1}{4} + 24} - 2\frac{1}{2} = \sqrt{30\frac{1}{4}} - 2\frac{1}{2} = 5\frac{1}{2} - 2\frac{1}{2} = 3$
$\frac{1}{2}x^2 + 10x = 56x^2 + 10x = 56x = \sqrt{25 + 56} - 5 = \sqrt{81} - 5 = 9 - 5 = 4$

$2^d cx^2 + a = bx$

$x^2 + 21 = 10x$

$x = \frac{10}{2} \pm \sqrt{(\frac{10}{2})^2 - 21} = 5 \pm \sqrt{25-21} = 5 \pm \sqrt{4} = 5 \pm 2 = 73$

$x^2 + a = bx(\frac{b}{2})^2 < a(\frac{b}{2})^2 = ax = \frac{b}{2}$

$(\frac{b}{2})^2 < a$

$cx^2 + a = bxx^2 + \frac{a}{c} = \frac{b}{c}x$

$3^d cx^2 = bx + a$

$x^2 = 3x + 4$

$x = \sqrt{(\frac{3}{2})^2 + 4} + \frac{3}{2} = \sqrt{2\frac{1}{4} + 4} + 1\frac{1}{2} = \sqrt{6\frac{1}{4}} + 1\frac{1}{2} = 2\frac{1}{2} + 1\frac{1}{2} = 4$

$ABABCGTKABDHDHAB$

	G	
C	AB	T
	K	

$ABABGDABABSHAB$

D	
AB	G

$ADADHNHBHCHCCDCHGGCHGGTCDGTCGGTTKKMMTTK$
$HBHBKTMTTAKMKLGKTGMLKLKMKGMRTAHTMRHB$
$MTMTHTMRKRRGGACGACCGCR$

AD
$HDADHCRDHBCHGHTGTAHTLGLAGKNTLGMAGMLAGGLCGNRMNTLHBKL$

$ARANKLAR$
$GMANKL$
$ADGCAGGMCGACAD$

$x^2+10x=39$
$4\times(\frac{b}{4})^2=(\frac{b}{2})^2$

$xyxy$

$x \pm ay \pm bxyxbayab$

$y \pm x = a \pm b$

$(+a)(+b) = +ab(-a)(-b) = +ab(+a)(-b) = -ab(-a)(+b) = -ab$

$(10+1) \times (10+2) = 10 \times 10 + 1 \times 10 + 2 \times 10 + 1 \times 2 = 100 + 10 + 20 + 2 = 132$

$(10-1) \times (10-1) = 10 \times 10 - 10 - 10 + 1 = 100 - 20 + 1 = 81$

$(10+2) \times (10-1) = 10 \times 10 - 10 + 20 - 2 = 100 - 10 + 20 - 2 = 108$

$(10-x) \times 10 = 10 \times 10 - 10x = 100 - 10x$

$(10+x) \times 10 = 10 \times 10 + 10x = 100 + 10x$

$(10+x)(10+x) = 10 \times 10 + 10x + 10x + x^2 = 100 + 20x + x^2$

$(10-x)(10+x) = 10 \times 10 - 10x + 10x - x^2 = 100 - x^2$

$\sqrt{200} - 10 + (20 - \sqrt{200}) = 10$
$20 - \sqrt{200} - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800}$
$50 + 10x - 2x^2 + (100 + x^2 - 20x) = 150 - 10x - x^2$
$100 + x^2 - 20x - [50 - 2x^2 + 10x] = 50 + 3x^2 - 30x$
$(2)^2 \times x^2 = 4x^2 \sqrt{4x^2} = 2x$
$3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9$
$3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9$
$\frac{1}{2}\sqrt{9} = \sqrt{\frac{1}{4} \times 9} = \sqrt{2\frac{1}{4}} = 1\frac{1}{2}$

$\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\frac{1}{2}$

$\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$

$\frac{2\sqrt{9}}{\sqrt{4}} = \frac{\sqrt{4} \times \sqrt{9}}{\sqrt{4}} = \sqrt{9} = 3$

$\sqrt{4} \times \sqrt{9} = \sqrt{4 \times 9} = \sqrt{36} = 6$

$\sqrt{10} \times \sqrt{5} = \sqrt{5 \times 10} = \sqrt{50}$

$\sqrt{\frac{1}{3}} \times \sqrt{\frac{1}{2}} = \sqrt{\frac{1}{3} \times \frac{1}{2}} = \sqrt{\frac{1}{6}}$

$3\sqrt{4} \times 2\sqrt{9} = \sqrt{9 \times 4} \times \sqrt{4 \times 9} = \sqrt{36 \times 36} = 36$

$ABACABCBBDACBHABHD$
$CBHDBHCBSHABBHACSBABCBBHSHSHHDBDACBSSD$

$ABACBDBHABCB$
$CBHDBSACSDSBBD$
$HDCBHGGDSDBHSBBCHGCBSD$

$(10-x)x = x^2 = 4(10-x)x = 40x - 5x^2 5x^2 = 40x x^2 = 8x x = 8(10-x) = 2$

$100 = x^2 \times 2\frac{7}{9} \frac{1}{2\frac{7}{9}} \times 100 = x^2 x = 6 10 - x = 4$

$2\frac{7}{9} = \frac{25}{9} \frac{1}{\frac{25}{9}} \times 100 = \frac{9}{25} \times 100 = 36 x^2 = 36 x = 6$

$\frac{10-x}{x} = 4 10 = 5x x = 2(10-x) = 8$

$(\frac{1}{3}x+1)(\frac{1}{4}x+1) = 20 \frac{1}{12}x^2 + \frac{7}{12}x + 1 = 20 x^2 + 7x = 228$

$(10-x)^2 + x^2 = 58 100 - 20x + x^2 + x^2 = 58 2x^2 - 20x + 100 = 58 x^2 - 10x + 50 = 29 x = 5 \pm \sqrt{25-29} = 5 \pm 2 = 7$

3

$\frac{1}{3}x \times \frac{1}{4}x = x + 24 \frac{1}{12}x^2 = x + 24 x^2 = 12x + 288 x = 6 + \sqrt{36+288} = 6 + 18 = 24$

$(10-x)x=2110x-x^2=21x=5\pm\sqrt{25-21}=5\pm 2$
$(10-x)^2-x^2=40100-20x+x^2-x^2=40100-20x=40100=20x+4060=20xx=3$
$100-20x+x^2+x^2+10-2x=54100-22x+2x^2=5455-11x+x^2=27x^2+28=11xx=\frac{11}{2}\pm\sqrt{\frac{121}{4}-28}=\frac{11}{2}\pm\frac{3}{2}=74$
$100+2x^2-20x=x(10-x)\times 2\frac{1}{6}=21\frac{2}{3}x-2\frac{1}{6}x^2100+4\frac{1}{6}x^2=41\frac{2}{3}x24+x^2=10xx=5-\sqrt{25-24}=5-1=4$
6
$4\frac{1}{6}=\frac{25}{6}\frac{1}{4\frac{1}{6}}=\frac{6}{25}=\frac{1}{5}+\frac{1}{5}\frac{1}{5}$
$\frac{a}{b}\times\frac{b}{a}=1$
$x^2+100=50xx=25-\sqrt{625-100}=25-\sqrt{525}$
$100-20x+x^2=81xx^2+100=101xx=\frac{101}{2}-\sqrt{\frac{10201}{4}-100}=50\frac{1}{2}-49\frac{1}{2}=1$
$mnrxrxmrx+nx=(m-n)+(rx-x)x=\frac{m-n}{mr+n-r+1}$

$\frac{x}{x+2}=\frac{1}{2}\frac{1}{2}(x+2)=x\frac{1}{2}x+1=x1=\frac{1}{2}xx=24$

$10x=(10-x)^2=100-20x+x^2$
$\frac{(10-x)x}{10-2x}=5\frac{1}{4}20\frac{1}{4}x=x^2+5\frac{1}{4}(10-2x)x=10-7\frac{1}{2}=2\frac{1}{2}$
$\frac{2}{3}\cdot\frac{1}{5}x^2=\frac{1}{7}x\frac{2}{15}x^2=\frac{1}{7}xx^2=\frac{15}{14}xx=\frac{15}{14}$
$\frac{3}{4}\cdot\frac{1}{5}x^2=\frac{4}{5}x\frac{3}{20}x^2=\frac{4}{5}x\frac{3}{20}x=\frac{4}{5}x=\frac{16}{3}=5\frac{1}{3}$

$4x^2=20x^2=5x=\sqrt{5}$

$x\times\frac{x}{3}=10x^2=30x=\sqrt{30}$

$x\times 4x=\frac{1}{3}x4x^2=\frac{1}{3}x12x^2=xx=\frac{1}{12}$
$x^2\times x=3x^2x^3=3x^2x=3$
$4x\times 3x=x^2+4412x^2=x^2+4411x^2=44x^2=4x=2$
$4x\times 5x=2x^2+3620x^2=2x^2+3618x^2=36x^2=2$
$x\times 4x=3x^2+504x^2=3x^2+50x^2=50x=\sqrt{50}$

$x^2 + 20 = 12xx = 6 \pm \sqrt{36 - 20} = 6 \pm 4 = 102$
$(x^2 - \frac{1}{3}x^2 - 3)^2 = x^2\frac{2}{3}x^2 - 3 = x\frac{4}{9}x^4 - 4x^2 + 9 = x^2x^2 + 9 = 5xx = 92\frac{1}{4}$

$\frac{1}{3}x \times \frac{1}{4}x = x\frac{1}{12}x^2 = xx^2 = 12xx = 12$

$\frac{x}{3} + 1 \times \frac{x}{4} + 2 = x + 13\frac{1}{12}x^2 + \frac{11}{12}x + 2 = x + 13\frac{1}{12}x^2 + 11 = \frac{1}{12}x + 13x^2 = 132 + xx = 12$
$xx^2 + x = \frac{3}{2}x = \sqrt{1\frac{1}{4} + \frac{1}{2}} - \frac{1}{2}$

$(x - \frac{1}{3}x - \frac{1}{4}x - 4)^2 = x + 12\frac{5}{12}x - 4x^2 + 23\frac{1}{4} = 24\frac{1}{4}x$

$x \times \frac{2}{3}x = 5\frac{2}{3}x^2 = 5x^2 = 7\frac{1}{2}x = \sqrt{7\frac{1}{2}}$

$\frac{x}{x+2} = \frac{1}{2}\frac{1}{2}(x+2) = x1 + \frac{1}{2}x = x1 = \frac{1}{2}xx = 24$

$\frac{1}{x} - \frac{1}{x+1} = \frac{1}{6}\frac{x+1-x}{x(x+1)} = \frac{1}{6}\frac{1}{x(x+1)} = \frac{1}{6}x^2 + x = 6x = \sqrt{\frac{1}{4} + 6} - \frac{1}{2} = 2$

$\frac{2}{3}x^2 = 5x^2 = 7\frac{1}{2}x = \sqrt{7\frac{1}{2}}$
$x^2 \times 3x = 5x^2 3x^3 = 5x^2 3x = 5x = 1\frac{2}{3}x^2 = 2\frac{7}{9}$
$(x^2 - \frac{1}{3}x^2) \times 3x = x^2\frac{2}{3}x^2 \times 3x = x^2 2x^3 = x^2 x = \frac{1}{2}x^2 = \frac{1}{4}$
$x^2 - 4x = \frac{1}{3}(x^2 - 4x)\frac{2}{3}(x^2 - 4x) = 4xx^2 - 4x = 6xx^2 = 10xx = 10x^2 = 100$
$\sqrt{x^2 - x} + x = 2\sqrt{x^2 - x} = 2 - xx^2 - x = 4 + x^2 - 4x3x = 4x = 1\frac{1}{3}x^2 = 1\frac{7}{9}$
$x^2 - 3x = xx^2 = 4xx = 4$

$abABa : b :: A : BaB = bA \therefore a = \frac{bA}{B} b = \frac{aB}{A}$

$ABCDACHRABTGABCDAKCKBKDKHTAKATHHKTATABAHACTHTRRGGH ADATAKHTATAHTH$

$1^{st} 3\frac{1}{7} d = p3.1428\, d 2^d \sqrt{10d^2} = p3.16227\, d 3^d\, \frac{d \times 62832}{20000} = p3.1416\, d$
$d\pi \frac{d^2}{4} = d^2 - \frac{d^2}{7} - \frac{1}{2} \cdot \frac{d^2}{7} = d^2(1 - \frac{3}{14}) = \frac{11}{14} d^2 \approx 0.7857 d^2\, \frac{22}{7} \cdot \frac{1}{4} = \frac{11}{14} \approx 0.7857$
$dd' s \frac{dd'}{2} = d \times \sqrt{s^2 - \frac{d^2}{4}}$
$\sqrt{10^2 - 5^2} = \sqrt{75} 5\sqrt{75} = 25\sqrt{5} 43.3+$
$\sqrt{169 - x^2} = \sqrt{29 + 28x - x^2} 169 = 29 + 28x 140 = 28x x = 5$

$nnx\frac{2}{3}[10+x] = 2x \therefore x = 5$

$\frac{4}{5}[10+x]-1 = 2x \therefore \frac{4}{5}[10+x]-1 = 2x\frac{2}{3}[10+x]-1 = x\frac{1}{5}[10+x]+1$

$\frac{1}{5}[10+x]-1$

$\frac{1}{8}\frac{1}{6}\frac{1}{8}+\frac{1}{6} = \frac{7}{24}\frac{17}{24}\frac{17}{48} = 1 = v1-\frac{1}{9} = \frac{8}{9} \therefore \frac{8}{9} = 48v \therefore v = \frac{1}{54} = \frac{1}{54}$

$\frac{1}{4}\frac{3}{4}\frac{1}{5}\frac{3}{4} = \frac{3}{20}\frac{6}{20}\frac{1}{8}+\frac{1}{7} = \frac{15}{56} = \frac{41}{56}\frac{1}{20} = \frac{41}{56}\times\frac{1}{20} = \frac{41}{1120}\frac{15}{56} = \frac{300}{1120} = \frac{15}{56}$

$\frac{1}{4}\frac{1}{6}\frac{1}{4}+\frac{1}{6} = \frac{5}{12}\frac{7}{12}\frac{2}{5}+\frac{1}{4} = \frac{13}{20}$

$\frac{1}{8}\frac{7}{8}x = \frac{1}{8}(1-x) = \frac{1}{4}\cdot\frac{7}{8}(1-x) = \frac{7}{32}-\frac{4}{32}(1-x) = x \therefore x = \frac{3}{35}$

$\frac{2}{7}x\frac{2}{7}(1-x) = x \therefore x = \frac{2}{9}1-x = \frac{7}{9} = \frac{2}{7}\cdot\frac{7}{9} = \frac{2}{9} = \frac{1}{7}\cdot\frac{7}{9} = \frac{1}{9} = \frac{2}{9}$

$x\frac{1}{8}\frac{7}{8}(1-x)\frac{1}{8}[1-x]\frac{1}{7}\cdot\frac{7}{8}(1-x)\frac{1}{8}\cdot\frac{8}{8}(1-x) = \frac{1}{7}\cdot\frac{7}{8}(1-x)-\frac{1}{8}(1-x) = x \therefore x = \frac{45}{336}$

$\frac{2}{13}\frac{1}{13}1-x-y = 13v1-[\frac{1}{5}-v]-[\frac{1}{4}-2v] = 13v1-\frac{1}{5}+v-\frac{1}{4}+2v = 13v$

$sdxd+x\frac{1}{3}[d+x]\frac{2}{3}[d+x]\frac{1}{3}[d+x] \therefore s-[d+x]+\frac{1}{3}[d+x]xs-\frac{2}{3}[d+x] = 2x \therefore \frac{3}{8}[3s-2d] = xs = 100 \therefore x = 35$ $d+x = 45\frac{1}{3}[d+x] = 152x = 70$

$pu\frac{2}{3}[u-p]\frac{1}{3}[u-p]\frac{1}{2}[u-p]\frac{2}{3}p\frac{1}{3}p\frac{1}{2}[u-p]\frac{1}{2}[u-p]$

$\frac{2}{3}\times\frac{1}{3}=\frac{2}{9}$

$auxu+x-a\frac{1}{3}[u+x-a]\frac{1}{2}[u+x-a]a-x\frac{1}{3}[u+x-a]\frac{1}{3}[(u+a-x)].\therefore\frac{1}{3}[(u+a-x)]=2x.\therefore x=\frac{1}{7}[u+a]u=a$ $x=\frac{2}{7}a{=}\ \frac{1}{7}[3u-2a]\frac{2}{7}[u+a]$

$=a{=}\ u{=}\ eha=300u=400e=10h=20{=}\ x{=}\ a-x{-}{=}\ u+x-a{-}{-}{=}\ u+x-a-e{=}\ \frac{1}{3}[u+x-a-e]$ $=\frac{2}{3}[u+x-a-e]\frac{1}{3}\times\frac{2}{3}[u+x-a-e]a-x+\frac{1}{3}[u+x-a-e]\frac{1}{3}[a-x]+\frac{2}{3}[u-e]2xx.\therefore x=\frac{3}{25}[7a+2[u-e]-3h]=108$

الخوارز□□ موسى بن محمد عبدالله ابو قال

المقابلةالجبر

xشيء

مال

واجبجذر

$c = \frac{62832}{20000}d\pi = 3.1416$